

Multilingual Corpora for Prosody and Code-Switching Research

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Project: *Prosody of Code-Switched Speech - Replication and Extension*
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1. Purpose and Context

We seek to replicate findings from the *Spanish-English Code-Switched Prosody* paper - which established that code-switched (CSW) utterances differ prosodically from monolingual ones and that variation is modulated by speaker proficiency - across other bilingual corpora.

The aim is to identify and prepare suitable multilingual datasets (e.g., Mandarin-English, Hindi-English) for applying a consistent DisVoice-based feature extraction pipeline, enabling cross-linguistic comparisons of pitch, energy, and duration trends.

2. Corpus Evaluation Framework

Criterion	Description
Prosody Suitability	Availability of naturalistic speech with clean audio and timestamps.
Language Identification (LID)	Token- or segment-level language labels.
Speaker Metadata	Demographic and proficiency indicators.
Accessibility	Licensing, file structure, and ease of integration into Python/Praat workflows.
Spontaneity	Conversational or natural dialogue vs. read speech.
Compatibility with DisVoice	Audio sampling (≥ 16 kHz mono), segment alignment, and utterance duration.

3. Comparison of Multilingual Code-Switching Corpora

Corpus	Language Pair(s)	Speech Style	Size / Speakers	Access	Prosody Relevance	DisVoice Readiness
Bangor Miami (BilingBank)	Spanish-English	Spontaneous dialogues	35 h / 84 spk	Open research license	Gold-standard for CS prosody with token-level LID	Yes - 16 kHz WAVs, easy segmentation
MaSaC	Hindi-English	Conversational + empathy tasks	40+ spk	Shared via Speech Lab	Indian bilingual prosody; proficiency metadata	Yes - needs transcript-to-timestamp alignment
MUCS (OpenSLR SLR104)	Hindi-English, Bengali-English	Lectures/tutorials (semi-spontaneous)	~140 h	Fully open (CC BY SA)	High volume; consistent annotation	Yes - native 16 kHz clips
ASCEND	Mandarin-English	Spontaneous dialogue	~10 h	Research-use repo	Clean, natural CS contexts	Yes - 16 kHz native
CS-Dialogue (2025)	Mandarin-English	Long-form spontaneous	104 h / 200 spk	Free for academic use	Ideal for cross-speaker mixed-effects analysis	Fully compatible
MCE (2024)	Cantonese-English	Real-world mixed speech	35 h	Contact authors	Tonal vs. non-tonal comparison	Likely compatible
HiACC (2025)	Hindi-English	Child + adult speech tasks	~25 h	Paper release pending	Developmental/proficiency analysis	Likely compatible

4. Observations and Recommendations

1. Best immediate extension:

- **MUCS** (Hinglish) for cross-validation with MaSaC.

- **ASCEND** (Mandarin–English) as a manageable, clean pilot for tonal influence on prosody.

2. High-impact future target:

- **CS-Dialogue** - modern, 104 h spontaneous corpus with balanced proficiency levels; best suited for scaling up replication.

3. DisVoice integration:

- All chosen corpora can be standardized to 16 kHz mono WAVs.
- Utterance-level slicing should follow timestamp or LID segmentation to ensure consistency with the Spanish–English baseline.

4. Thematic relevance:

- Enables testing if **prosodic convergence toward the dominant language** generalizes across **non-Indo-European and tonal languages**, strengthening cross-linguistic theory.

5. Integration Plan

Stage	Action	Corpus	Expected Output
1. Pilot Replication (Fall 2025)	Reproduce Spanish-English contrasts	MaSaC	Feature CSV + plots
2. Extension (Winter 2026)	Apply pipeline to MUCS (HI-EN) and ASCEND (ZH-EN)	MUCS / ASCEND	Contrast & effect-size tables
3. Modeling (Spring 2026)	Mixed-effects modeling of proficiency × prosody	CS-Dialogue	Cross-corpus trends
4. Dissemination (Summer 2026)	Prepare paper submission	All	Speech Prosody / Interspeech draft

6. Current Progress

- Environment (`csw-prosody`) and dependencies set up locally.
 - MaSaC dataset unzipped and organized under `data/masac_raw/`.
 - Transcript mapping in progress; DisVoice extraction pilot scheduled.
 - Reviewed and compiled corpus comparison table (this document).
 - Plan to finalize the formatted version and discuss selection at **Wednesday's meeting**.
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7. Next Steps

1. Finalize MaSaC manifest and run first-pass DisVoice extraction.
 2. Confirm MUCS and CS-Dialogue access permissions.
 3. Run comparative feature summary (pitch / energy / duration).
 4. Prepare plots for proficiency × language interaction trends.
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